



UNIVERSITAT ROVIRA I VIRGILI (URV) Y UNIVERSITAT OBERTA DE CATALUNYA (UOC)

MASTER IN COMPUTATIONAL AND MATHEMATICAL ENGINEERING

FINAL MASTER PROJECT

AREA: YYY

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Barcelona, November 20, 2025

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FINAL PROJECT SHEET

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Program:	Master in Computational and Mathematical Engineering
Area:	Name of the FMP area
Language:	English
Key words	Maxim 3 key words

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Acknowledgments

If deemed appropriate, mention the persons, companies or institutions that have contributed to the realization of this project.

Abstract

Text with the synthesis of the project, that is, a text in which the definition of the project / problem addressed, its objectives / methods of resolution, and the results and conclusions are briefly explained (it can not be a list, but a text continuous written in a structured way). If it is necessary to put a reference in this text, it will be annotated at the bottom of the same page. In this section you can use a more literary and colloquial language than for the rest of the document.

In case of not writing the the document in English, it will be necessary to write the Abstract in duplicate. One of the versions must be textbf obligatorily in English. The other version can be written in Catalan or Spanish, in the language used for the rest of the report. The word Abstract will be changed to “ textbf Resum ” or “ textbf Resumen ” in the Catalan and Spanish versions, respectively.

Recommended length: 250 words maximum.

How to write a good abstract (in English):

<http://www.ece.cmu.edu/koopman/essays/abstract.html>

Keywords: Work keywords separated by commas. For example for this document they could be Model, Guideline, Template, Memory, Final Master Project / Master

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Chapter 1

Introducción

1.1 General description of the problem

At present, data mining processes require large amounts of data, which in many cases contain personal and private information of users or people. Although basic anonymization processes are carried out on the data, that is, elimination of names or other key identifiers, there are a multitude of re-identification techniques that allow to re-identify a user within this data set. Figure [ref fig: context-anoni1](#) presents a map where it is possible to contextualize the processes of anonymization and re-identification within a process of data mining.

1.1.1 Example of subsection

Although important advances have been made in the preservation of privacy in the publication of data, such as the model k -anonymity [\[1\]](#).

An example of a pseudo-code can be found in the [Code 1](#)

An example of a table can be seen in the [Table 1.1](#)

Node ID	$\mathcal{H}_0$	$\mathcal{H}_1$	$\mathcal{H}_2$
Alice	ϵ	1	{4}
Bob	ϵ	4	{1, 1, 4, 4}
Carol	ϵ	1	{4}
Dave	ϵ	4	{2, 4, 4, 4}
Ed	ϵ	4	{2, 4, 4, 4}
Fred	ϵ	2	{4, 4}
Greg	ϵ	4	{2, 2, 4, 4}
Harry	ϵ	2	{4, 4}

Table 1.1: *Vertex refinement queries.*



Figure 1.1: Foot of the image.

Algorithm 1 Pseudocode of the algorithm *Random Switch*

Input: The original graph G and the percentage of anonymization p that you want to apply.

Output: El graph G anonymized

```

num = round( $G.num\_edges()$  *  $p$ )
i = 0
while i < num do
    e1 =  $G.random\_edge()$ 
    e2 =  $G.random\_edge()$ 
    new_e1 = (e1.origen, e2.origen)
    new_e2 = (e1.destino, e2.destino)
    if ! $G.exist(new\_e_1)$  and ! $G.exist(new\_e_2)$  then
         $G.add\_edge(new\_e_1)$ 
         $G.add\_edge(new\_e_2)$ 
         $G.delete\_edge(e_1)$ 
         $G.delete\_edge(e_2)$ 
        i = i + 1
    end if
end while
return  $G$ 

```

Bibliography

- [1] Latanya Sweeney. k-anonymity: a model for protecting privacy. *Int. J. Uncertain. Fuzziness Knowl.-Based Syst.*, 10:557–570, October 2002.